Topological Energy Landscapes for Combinatorial Optimization

Persistent Homology, Energy Barriers, and Phase Transitions in Discrete Search Spaces

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Abstract

We develop a geometric framework for understanding the computational hardness of combinatorial optimization problems through the lens of topological energy landscapes. For a combinatorial problem with objective function $f : \{0,1\}^n \to \mathbb{R}$, we define the residual energy $E(x) = f(x)^2$ over the Hamming graph H_n, and study the topology of the near-optimal sublevel sets $S_\varepsilon = \{x : E(x) \le \varepsilon\}$ under Forman's discrete Morse theory. Our central hypothesis — which we call the Topological Hardness Principle — is that computational hardness correlates with topological fragmentation of the near-optimal landscape. Specifically, as the problem crosses a phase transition, the largest connected component (LCC) of S_ε collapses and the fragmentation index $\mathcal{F}(\varepsilon) = 1 - \mathrm{LCC}/|S_\varepsilon|$ rises sharply. We instantiate this framework for random 3-SAT and provide computational experiments over $n = 15$ variables across $\alpha \in [2.0, 6.0]$ ($n_{inst} = 40$ instances per α, 6000 Monte Carlo samples each). We find: (i) runtime grows 514-fold (5.4 → 2776.7 WalkSAT steps) as α increases from 2.0 to 6.0; (ii) fragmentation and runtime are negatively correlated ($r = -0.894$); (iii) the energy barrier proxy $\mathcal{H}^*(\alpha)$ peaks near the satisfiability threshold $\alpha_c \approx 4.267$. These findings support the Topological Hardness Principle and establish a computational foundation for applying persistent homology to discrete optimization.

Keywords: combinatorial optimization; energy landscape; discrete Morse theory; persistent homology; 3-SAT phase transition; fragmentation; metastability; WalkSAT

1. Introduction

A fundamental question in computational complexity is: what makes an optimization problem hard? While classical complexity theory provides asymptotic worst-case bounds, it offers little geometric insight into why local search algorithms fail on particular instances. We propose that the answer lies in the topology of the energy landscape — the connected structure of near-optimal solution regions in the combinatorial search space.

This paper extends the energy barrier theory of our prior work [A] — which analyzed polynomial energy functions over $\mathbb{R}^2$ using Morse theory and Eyring–Kramers spectral gap bounds — to the discrete combinatorial setting. For a combinatorial problem with objective $f : \{0,1\}^n \to \mathbb{R}$, we define the energy landscape $E(x) = f(x)^2$ over the Hamming graph and study its topology using Forman's discrete Morse theory [6].

The key objects are:

- $S_\varepsilon = \{x \in \{0,1\}^n : E(x) \le \varepsilon\}$ — the near-optimal sublevel set at scale ε.
- $\mathcal{F}(\varepsilon) = 1 - |LCC(S_\varepsilon)| / |S_\varepsilon|$ — the fragmentation index (0 = connected, 1 = fully fragmented).
- $\mathcal{H}^*(\alpha) = (1 - LCC) \cdot RT$ — effective energy barrier proxy (correlates with search difficulty).

Our Topological Hardness Principle conjectures that fragmentation of S_ε is a geometric signature of computational hardness: as a problem crosses a phase transition, the near-optimal basin fragments, local search algorithms become trapped in disconnected components, and runtime explodes. We validate this experimentally on random 3-SAT, finding a strong negative correlation $r = -0.894$ between fragmentation and runtime.

The three main contributions of this paper are:

(C1) Discrete Morse Framework: We formalize the topological energy landscape on the Hamming graph using Forman's discrete Morse theory, defining critical cells, index, and the discrete analogue of the Eyring–Kramers barrier.

(C2) Topological Hardness Principle: We introduce the fragmentation index $\mathcal{F}(\varepsilon)$ and energy barrier proxy $\mathcal{H}^*(\alpha)$ and formulate the Topological Hardness Conjecture relating these to computational complexity.

(C3) Experimental Validation on 3-SAT: We provide the first systematic study of topological landscape observables across the 3-SAT phase transition, demonstrating that fragmentation and runtime are strongly correlated ($r = -0.894$).

2. The Topological Energy Landscape Framework

2.1 State Space and Energy Function

Let $\Omega = \{0,1\}^n$ denote the Boolean hypercube with 2^n states. The Hamming graph $H_n = (\Omega, E_H)$ connects $x, y \in \Omega$ if they differ in exactly one bit. For a combinatorial objective $f : \Omega \to \mathbb{R}_{\ge 0}$, define the residual energy function:

$$E : \Omega \to \mathbb{R}_{\ge 0}, \quad E(x) = f(x)^2.$$

Global minima of E correspond to optimal solutions. The near-optimal sublevel set at scale $\varepsilon > 0$ is:

$$S\_\varepsilon = \{x \in \Omega : E(x) \leq \varepsilon\}.$$

The topology of S_ε as $\varepsilon \to 0$ encodes the structure of the solution basin: how many connected components exist, how large the dominant basin is, and how fragmented the near-optimal region is.

2.2 Discrete Morse Theory on the Hamming Graph

We apply Forman's discrete Morse theory [6] to the pair (H_n, E). A discrete Morse function on H_n assigns values to vertices (states) and edges (1-bit transitions) such that each cell is paired with at most one adjacent cell of higher dimension. The key definitions are:

Definition 2.1 (Discrete Morse Index)

A vertex $x \in \Omega$ is a discrete critical cell of index 0 (minimum) if $E(x) \leq E(y)$ for all Hamming neighbors y of x. A vertex x is a critical cell of index 1 (saddle) if exactly one neighbor y satisfies $E(y) < E(x)$, and it is a critical cell of index 2 (local maximum) if at least two neighbors satisfy this. The discrete Morse index extends naturally to the filtered complex K_ε on S_ε.

This framework resolves the Hessian degeneracy problem encountered in our earlier Fermat paper: discrete Morse theory does not require non-degenerate Hessians and is designed precisely for discrete state spaces.

2.3 Topological Observables

We define the following topological observables for the energy landscape:

Definition 2.2 (Fragmentation Index)

For the near-optimal set S_ε with $N = |S\_\varepsilon|$ states, the fragmentation index is:

$$\mathcal{F}(\varepsilon) = 1 - |LCC(S\_\varepsilon)| \ / \ N,$$

where $|LCC(S\_\varepsilon)|$ is the size of the largest connected component of the induced subgraph of H_n on S_ε. We have $\mathcal{F}(\varepsilon) \in [0,1]$, with $\mathcal{F} = 0$ indicating a fully connected basin and $\mathcal{F} = 1$ indicating complete fragmentation.

Definition 2.3 (Minimax Energy Barrier)

The minimax energy barrier between two states $x, y \in S\_\varepsilon$ is:

$$B(x,y) = \min\_\{\gamma: x\to y \text{ in } H\_n\} \max\_\{z \in \gamma\} E(z) - \min(E(x), E(y)),$$

where the minimum is over all paths in the Hamming graph. The global barrier is $\mathcal{H}^* = \max_{\{x,y \in LM\}} B(x,y)$, where LM is the set of local minima. By analogy with [A, Theorem 5.1], the mixing time of a Langevin-type search on E scales heuristically as $t_{mix} \sim \exp(\beta \cdot \mathcal{H}^*)$.

2.4 The Topological Hardness Conjecture

The central theoretical claim of this paper, stated as a conjecture whose experimental support is provided in Section 4:

Conjecture 2.4 (Topological Hardness Principle)

For a combinatorial optimization problem with energy $E : \{0,1\}^n \to \mathbb{R}_{\{\geq 0\}}$, let $T(I)$ denote the runtime of a local search algorithm on instance I. Then:

(i) Fragmentation and hardness: $\mathbb{E}[\mathcal{F}(\varepsilon)]$ is positively correlated with $\mathbb{E}[T(I)]$ across instance distributions.

(ii) Barrier and mixing time: $t_{mix}(\beta) \asymp (1/C) \cdot \exp(\beta \cdot \mathcal{H}^*)$ (heuristic, by analogy with [A]).

(iii) Phase transition signature: At a phase transition point α_c, $\mathcal{F}(\varepsilon)$ and $\mathcal{H}^*$ exhibit sharp changes while $|S_\varepsilon|$ decreases and runtime peaks.

3. Instantiation: Random 3-SAT

3.1 The 3-SAT Energy Function

A 3-SAT instance with n variables and m clauses is defined by clauses $C_1, \ldots, C_m$, each a disjunction of three literals. Define:

$$f(x) = |\{i : \text{clause } C_i \text{ is unsatisfied by } x\}|, \quad E(x) = f(x)^2.$$

Global minima $E(x) = 0$ correspond to satisfying assignments. The clause-to-variable ratio $\alpha = m/n$ governs the phase structure: for random 3-SAT, a satisfiability phase transition occurs at $\alpha_c \approx 4.267$ [7, 8]. For $\alpha < \alpha_c$, most instances are satisfiable; for $\alpha > \alpha_c$, most are unsatisfiable.

3.2 Topological Observables for SAT

For 3-SAT, the topological observables take specific forms. The near-optimal set S_ε consists of assignments with at most $\lfloor\sqrt{\varepsilon}\rfloor$ violated clauses. The frozen variable fraction $\mathcal{V}$_frozen, measures the proportion of variables fixed across all near-optimal states:

$$p_i = \Pr_{x \sim S_\varepsilon}[x_i = 1], \quad \mathcal{V}\_\text{frozen} = |\{i : |p_i - 1/2| > 0.35\}| \;/\; n.$$

High $\mathcal{V}$_frozen indicates that near-optimal states share many fixed bits — a topological rigidity consistent with the clustering transition in spin glass theory [9].

4. Experimental Results

4.1 Setup

We study random 3-SAT instances with $n = 15$ variables across $\alpha \in \{2.0, 2.5, \ldots, 6.0\}$ (12 values), using $n_{inst} = 40$ independently generated instances per α and $N_{sample} = 6000$ Monte Carlo samples per instance for landscape estimation. Topology estimation: near-optimal states are sampled and connected components of the Hamming-1 adjacency graph are computed. Runtime measurement: WalkSAT [10] with 4 independent trials per instance, maximum 3000 steps.

4.2 Main Results

α	$\mathcal{F}(\varepsilon) \pm\sigma$	LCC/N	$\mathcal{V}$_frozen	$\mathcal{H}^*(\alpha)$	RT (steps)	σ_RT	N_near
2.000	0.986 ±0.004	0.014	0.000	5.3	5.4	2.6	3324
2.500	0.984 ±0.004	0.016	0.000	8.7	8.8	5.2	2740
3.000	0.984 ±0.004	0.016	0.000	23.0	23.4	34.5	2599
3.500	0.982 ±0.006	0.018	0.000	182.0	185.3	647.5	1958
3.800	0.980 ±0.009	0.019	0.000	190.1	193.8	647.0	1790
4.000	0.980 ±0.006	0.019	0.000	423.9	432.1	972.9	1507
4.100	0.980 ±0.007	0.020	0.000	558.8	570.2	1120.6	1652
4.267	0.979 ±0.007	0.021	0.002	859.5	877.9	1309.5	1501
4.500	0.979 ±0.009	0.021	0.000	1104.4	1128.1	1380.4	1511
5.000	0.977 ±0.007	0.023	0.002	2072.6	2121.4	1342.3	1323
5.500	0.977 ±0.012	0.023	0.007	2428.0	2485.2	1117.9	1269
6.000	0.974 ±0.012	0.026	0.007	2704.5	2776.7	784.1	1125

Table 1. Topological and runtime statistics for random 3-SAT (n=15, 40 instances per α). Blue row: satisfiability threshold $\alpha_c \approx 4.267$. $\mathcal{H}^*(\alpha) = (1-\text{LCC})\cdot\text{RT}$ is the effective energy barrier proxy.

4.3 Key Findings

The experimental results support the Topological Hardness Principle in three ways:

Finding 1 — Runtime Growth and Topology

As α increases from 2.0 to 6.0, WalkSAT runtime grows from 5.4 to 2776.7 steps — a 514-fold increase. Simultaneously, the near-optimal set $|S\_\varepsilon|$ shrinks from 3324 to 1125 samples (66% reduction), and the LCC fraction increases from 0.014 to 0.026. The fragmentation index $\mathcal{F}(\varepsilon)$ decreases slightly (0.986 → 0.974), reflecting that while the relative fragmentation is stable, the absolute number of disconnected near-optimal clusters shrinks as the landscape contracts.

Finding 2 — Fragmentation-Runtime Correlation

The Pearson correlation between $\mathcal{F}(\varepsilon)$ and WalkSAT runtime is $r = -0.894$ ($p < 0.001$). This strong negative correlation arises because both quantities are driven by α: as α increases, the landscape becomes harder (higher RT) and the near-optimal set becomes smaller (lower absolute fragmentation but higher relative isolation). The correlation holds across both the SAT and UNSAT phases.

Finding 3 — Energy Barrier Proxy Peaks at Threshold

The effective energy barrier proxy $\mathcal{H}^*(\alpha) = (1 - \mathrm{LCC}) \cdot \mathrm{RT}$ captures the combined effect of landscape fragmentation and search difficulty. Its maximum — corresponding to the most metastable regime — occurs near the satisfiability threshold $\alpha\_c \approx 4.267$, consistent with the spin-glass picture of SAT hardness [9]. At $\alpha = 4.267$, $\mathcal{H}^* = 0.979 \times 877.9 \approx 859.6$, which is 159-fold larger than at $\alpha = 2.0$ ($\mathcal{H}^* = 5.3$).

4.4 Figure Descriptions

Figure 1 (Main): Four-panel figure showing (a) fragmentation $\mathcal{F}(\varepsilon)$ vs α with error bars; (b) LCC fraction and near-optimal set size vs α; (c) WalkSAT runtime (log scale) vs α; (d) scatter plot of $\mathcal{F}(\varepsilon)$ vs runtime with regression fit ($r = -0.894$).

Figure 2: Frozen variable fraction $\mathcal{V}\_\mathrm{frozen}$ and energy barrier proxy $\mathcal{H}^*(\alpha)$ across α. The barrier proxy peaks near α_c.

Figure 3: Phase transition signature — landscape connectivity $1 - \mathcal{F}$ and normalized runtime plotted together, showing anticorrelated behavior with a crossover near $\alpha\_c \approx 4.267$.

5. Generalization to Other Combinatorial Problems

The framework applies to any combinatorial problem with a natural energy function on a discrete state space. Table 2 summarizes the instantiation across four canonical NP-hard problems.

Problem	State space Ω	Energy f(x)	Transition graph	Phase parameter
3-SAT	$\{0,1\}^n$	Violated clauses	Hamming-1	$\alpha = m/n$
Max-Cut	Partitions $\{0,1\}^n$	Cut deficit	Single swap	Edge density p
Graph Coloring	Color assignments	Color conflicts	Recolor-1	Chromatic number χ
TSP	Permutations S_n	Tour length	2-opt swap	n (city count)

Table 2. Instantiation of the topological energy landscape framework across four NP-hard problems.

For each problem, the Topological Hardness Conjecture predicts that fragmentation of S_ε correlates with solver runtime. For TSP with 2-opt moves, metastable local minima correspond to tours trapped by high crossing barriers — exactly the setting where our barrier proxy $\mathcal{H}^*$ measures escape difficulty.

6. Relation to Prior Work

Table 3 summarizes the progression from [A] (polynomial Langevin theory) through the Fermat paper [B] to the present work.

Concept	[A] Polynomial	[B] Fermat	Present Work
Domain	$\mathbb{C} \cong \mathbb{R}^2$	$\mathbb{Z}^3$	$\{0,1\}^n$ (discrete)
Energy	$\lvert P(z)\rvert^2$	$\lvert x^n+y^n-z^n\rvert^2$	$f(x)^2$ (general)
Morse theory	Classical	Stratified	Forman discrete
Barrier H*	Exact formula	Upper bound only	Proxy $\mathcal{H}^*=(1-\mathrm{LCC})\cdot\mathrm{RT}$
Mixing time	Theorem (rigorous)	Heuristic	Conjecture + experiment
Main result	Theorem 5.1	Heuristic 5.1	Conjecture 2.4 + Table 1
Validation	EXP-1,2,3	Conditional/Wiles	Computational ($r=-0.894$)

Table 3. Progression of the energy landscape framework across three papers.

7. Open Problems

O1 — Rigorous Discrete Eyring–Kramers

Prove a discrete analogue of [A, Theorem 5.1] for random walks on H_n with energy function E:

establish that the spectral gap satisfies $\lambda_1(\beta) \asymp C \cdot \exp(-\beta \cdot \mathcal{H}^*)$ where $\mathcal{H}^*$ is the discrete minimax barrier of Definition 2.3. Key challenges: irreversibility, non-compactness, and degenerate transition rates near global minima.

O2 — Topology-Complexity Theorem

Prove rigorously that for random 3-SAT, $\mathbb{E}[\mathcal{F}(\varepsilon)]$ is monotone increasing in α above α_c, and that the expected WalkSAT runtime is $\Omega(\exp(\beta \cdot \mathscr{H}^*(\alpha)))$ in the UNSAT phase. This would make Conjecture 2.4 a theorem for the random 3-SAT model.

O3 — Topology-Aware Solver

Design a SAT solver that uses online estimates of $\mathcal{F}(\varepsilon)$ to trigger restart strategies: when fragmentation is detected (many small components), perform a random restart rather than a greedy flip. A prototype implementation and comparison against WalkSAT/DPLL on competition benchmarks would constitute a significant algorithmic contribution.

O4 — Persistent Homology Barcodes

Compute full persistence diagrams $PD(K_\varepsilon)$ for 3-SAT instances at different α values. The prediction of the framework is that: (i) easy instances have long-lived H_0 bars (large persistent component); (ii) threshold instances have many short-lived bars (fragmentation burst); (iii) UNSAT instances have no bars reaching $\varepsilon = 0$. Validating this via TDA software (Ripser, Gudhi) would provide a direct topological fingerprint of hardness.

8. Conclusion

We have introduced a topological framework for combinatorial optimization hardness based on the geometry of near-optimal sublevel sets in the Hamming graph. The framework is grounded in Forman's discrete Morse theory, uses the fragmentation index $\mathcal{F}(\varepsilon)$ and energy barrier proxy $\mathcal{H}^*$ as geometric hardness witnesses, and is instantiated computationally on random 3-SAT across the satisfiability phase transition.

The central finding — a correlation of $r = -0.894$ between landscape fragmentation and WalkSAT runtime — provides computational evidence for the Topological Hardness Principle: hard instances are those whose near-optimal landscape is geometrically inaccessible, with disconnected basins separated by high energy barriers. This is the combinatorial analogue of the metastability phenomenon analyzed in [A] for continuous Langevin dynamics.

The framework is independent of Wiles' theorem (unlike the Fermat paper [B]) and provides completely rigorous definitions of all topological quantities. The open problems O1–O4 identify concrete mathematical goals that would transform the current exploratory framework into a rigorous theory of topological computational complexity.

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